

Practice problems #11

1. In the following, assume that T is a linear transformation. The vectors e_1 and e_2 are the columns of the 2×2 identity matrix. Find the standard matrix of T :

(a) $T : \mathbf{R}^2 \rightarrow \mathbf{R}^4$, $T(e_1) = (1, 2, 0, 5)$ and $T(e_2) = (3, -6, 1, 0)$. Fill in the blank and state why: This transformation is _____.

(b) $T : \mathbf{R}^2 \rightarrow \mathbf{R}^2$ is a transformation that maps e_1 into $e_1 + 2e_2$ but leaves the vector e_2 unchanged. Fill in the blank and state why: This transformation is _____.

2. Let

$$A = \begin{bmatrix} 1 & -3 \\ -2 & 4 \end{bmatrix}, \quad \text{and} \quad x = \begin{bmatrix} 5 \\ 3 \end{bmatrix}.$$

Compute $(Ax)^T$ and $x^T x$.

3. Produce a 3×3 symmetric matrix $A = [a_{ij}]$ with $a_{13} \neq 0$.
4. An $n \times n$ matrix is said to be *skew-symmetric* if $A = -A^T$. Produce a 2×2 skew-symmetric matrix $A = [a_{ij}]$ with $a_{21} \neq 0$.

Answers or Hints on page 2.

#1(a): $A = \begin{bmatrix} 1 & 3 \\ 2 & -6 \\ 0 & 1 \\ 5 & 0 \end{bmatrix}$, one-to-one. State why.

#1(b): What are $T(e_1)$ and $T(e_2)$? Ans: $A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$, one-to-one and onto. State why.

#2: $(Ax)^T$ has 2 in the answer, and $x^T x$ has 4 in the answer if you calculate them correctly.

#3: What is the definition of a symmetric matrix?

#4: Do it like #2.